$$C_\phi^{(11)\ (22)\ (22)} \to$$

$$\frac{1}{16\pi^2}\ \Big(\frac{1}{18}(-g_1^2 g_2^4+g_2^6)\,\mathrm{LF}_{3,0}[m_2]+\frac{1}{12}(-g_1^2 g_2^4+g_2^6)\,\mathrm{LF}_{4,-1}[m_2]+\frac{4}{45}g_2^4(g_1^2-g_2^2)\,\mathrm{LF}_{5,-2}[m_2]+$$

$$\frac{1}{72}\sum_p g_1^6\,\mathrm{LF}_{3,0}[m_{\tilde{d}}^{p}]-\frac{1}{6}g_1^4\,\overline{y_d}^{pr}y_d^{pr}\,\mathrm{LF}_{3,0}[m_{\tilde{d}}^{r}]+$$

$$\frac{1}{8}\sum_p g_1^6\,\mathrm{LF}_{3,0}[m_{\tilde{e}}^{p}]-\frac{1}{2}g_1^4\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{3,0}[m_{\tilde{e}}^{r}]-$$

$$\frac{1}{288}(g_1^2-g_2^2)\left(36\,\overline{y_e}^{pr}y_e^{pr}(g_1^2+3g_2^2)+\sum_p(9g_1^4+9g_1^2g_2^2-10g_2^4)\right)\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]+$$

$$\frac{1}{96}g_2^2(-g_1^2+g_2^2)\left(-24\,\overline{y_e}^{pr}y_e^{pr}+\sum_p g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]+$$

$$\frac{1}{45}\sum_p g_2^4(-g_1^2+g_2^2)\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]+\frac{1}{288}\Big(-12\,\overline{y_d}^{pr}y_d^{pr}(g_1^4+18g_1^2g_2^2-27g_2^4)+$$

$$6\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+3g_2^2)^2+\sum_p(g_1^6+21g_1^2g_2^4-30g_2^6)\Big)\mathrm{LF}_{3,0}[m_{\tilde{q}}^{p}]+$$

$$\frac{1}{32}g_2^2(-g_1^2+g_2^2)\left(-24\,\overline{y_d}^{pr}y_d^{pr}+\sum_p g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{q}}^{p}]+\frac{1}{15}\sum_p g_2^4(-g_1^2+g_2^2)\,\mathrm{LF}_{5,-2}[m_{\tilde{q}}^{p}]-$$

$$\frac{1}{9}\sum_p g_1^6\,\mathrm{LF}_{3,0}[m_{\tilde{u}}^{p}]+\frac{1}{3}g_1^4\,\overline{y_u}^{pr}y_u^{pr}\,\mathrm{LF}_{3,0}[m_{\tilde{u}}^{r}]+\frac{1}{36}(-g_1^2g_2^4+g_2^6)\,\mathrm{LF}_{3,0}[\tilde{\mu}]+$$

$$\frac{1}{24}(-g_1^2g_2^4+g_2^6)\,\mathrm{LF}_{4,-1}[\tilde{\mu}]+\frac{2}{45}g_2^4(g_1^2-g_2^2)\,\mathrm{LF}_{5,-2}[\tilde{\mu}]-$$

$$\frac{1}{4}g_1^6\,\mathrm{LF}_{3,3,-3}[m_1,\tilde{\mu}]-\frac{3}{4}g_1^6m_1^2\,\mathrm{LF}_{3,3,-2}[m_1,\tilde{\mu}]+\frac{1}{3}g_2^4(g_1^2-g_2^2)\,\mathrm{LF}_{2,1,0}[m_2,\tilde{\mu}]+$$

$$\frac{17}{24}g_2^4(g_1^2-g_2^2)\,\mathrm{LF}_{2,2,-1}[m_2,\tilde{\mu}]+\frac{1}{6}(-g_1^2g_2^4+g_2^6)\,\mathrm{LF}_{3,1,-1}[m_2,\tilde{\mu}]+$$

$$\frac{7}{6}g_2^4(-g_1^2+g_2^2)\,\mathrm{LF}_{3,2,-2}[m_2,\tilde{\mu}]+\frac{3}{4}g_2^4m_2^2(g_1^2-g_2^2)\,\mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}]+$$

$$\frac{1}{4}g_2^4(2g_1^2-11g_2^2)\,\mathrm{LF}_{3,3,-3}[m_2,\tilde{\mu}]-\frac{1}{4}g_2^4m_2^2(2g_1^2+g_2^2)\,\mathrm{LF}_{3,3,-2}[m_2,\tilde{\mu}]+$$

$$\frac{1}{2}g_2^4(g_1^2-g_2^2)\,\mathrm{LF}_{4,2,-3}[m_2,\tilde{\mu}]+\frac{1}{2}g_2^4m_2^2(-g_1^2+g_2^2)\,\mathrm{LF}_{4,2,-2}[m_2,\tilde{\mu}]+$$

$$\frac{1}{2}g_1^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]-\frac{1}{12}g_1^4\left(4\,\overline{a_d}^{pr}a_d^{pr}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\right)\mathrm{LF}_{2,2,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{12}g_1^4\left(-2\,\overline{a_d}^{pr}a_d^{pr}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\right)\mathrm{LF}_{3,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$

$$\frac{1}{2}g_1^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{2,1,0}[m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+\frac{1}{2}g_1^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$

$$\frac{1}{4}g_1^4\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\,\mathrm{LF}_{2,2,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+\frac{1}{4}g_1^4\left(-2\,\overline{a_e}^{pr}a_e^{pr}+\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\right)\mathrm{LF}_{3,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$

$$\frac{1}{4}\left(2g_2^4\,\overline{a_e}^{pr}a_e^{pr}+g_1^4\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}\right)\mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$

$$\frac{1}{8}\left(2g_1^4\,\overline{a_e}^{pr}a_e^{pr}+g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}(g_1^2-g_2^2)\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$

$$\frac{1}{8}\left(-2g_2^2\,\overline{a_e}^{pr}a_e^{pr}(g_1^2+2g_2^2)+3g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}(-g_1^2+g_2^2)\right)\mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$\frac{1}{24}(g_1^2-g_2^2)\left(2\,\overline{a_e}^{pr}a_e^{pr}(3g_1^2+g_2^2)+3\tilde{\mu}^2\,\overline{y_e}^{pr}y_e^{pr}(-g_1^2+g_2^2)\right)\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$\frac{1}{2}g_2^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$

$$\frac{1}{4}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$

$$\frac{1}{12}\left(2\,\overline{a_d}^{pr}a_d^{pr}(-4g_1^4+9g_2^4)+g_1^4\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}\right)\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$

$$\frac{1}{8}\left(2g_1^4\,\overline{a_d}^{pr}a_d^{pr}+g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}(g_1^2-g_2^2)\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$

$$\frac{3}{8}\left(2\,\overline{a_d}^{pr}a_d^{pr}(2g_1^4-g_1^2g_2^2-2g_2^4)+g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}(-g_1^2+g_2^2)\right)\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$

$$\frac{1}{8}(g_1^2-g_2^2)\left(2\,\overline{a_d}^{pr}a_d^{pr}(3g_1^2+g_2^2)+3\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{pr}(-g_1^2+g_2^2)\right)\mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$

$$\frac{1}{4}\,\overline{y_d}^{pr}y_d^{sr}\left(2\,\overline{y_d}^{st}y_d^{pt}(2g_1^2-3g_2^2)-\overline{y_u}^{st}y_u^{pt}(g_1^2+3g_2^2)\right)\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{4}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{48}\left(\overline{a_u}^{pr}a_u^{pr}(-7g_1^4-18g_1^2g_2^2+9g_2^4)+2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^4-6g_1^2g_2^2-3g_2^4)\right)$$

$$\mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\frac{1}{48}\left(\overline{a_u}^{pr}a_u^{pr}(g_1^2+3g_2^2)^2-2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^4-18g_1^2g_2^2+9g_2^4)\right)$$

$$\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\frac{1}{4}g_2^2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+\frac{3}{4}g_2^2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}$$

$$(-g_1^2+g_2^2)\,\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-\frac{1}{4}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{48}\left(\overline{a_u}^{pr}a_u^{pr}(7g_1^2-3g_2^2)^2-2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^4-6g_1^2g_2^2-3g_2^4)\right)\mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{16}\left(\overline{a_u}^{pr}a_u^{pr}(g_1^4+2g_1^2g_2^2-3g_2^4)+2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)^2\right)\mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$

$$\frac{3}{16}\left(\overline{a_u}^{pr}a_u^{pr}(7g_1^4-10g_1^2g_2^2+3g_2^4)+2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(g_1^2+g_2^2)^2\right)\mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{8}(g_1^2-g_2^2)\left(3\,\overline{a_u}^{pr}a_u^{pr}(g_1^2-g_2^2)-2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pr}(3g_1^2+g_2^2)\right)\mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{2}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(2g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+$$

$$\frac{3}{2}g_2^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+\frac{1}{8}(g_1^6+2g_1^4g_2^2-3g_1^2g_2^4)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_1]-$$

$$\frac{3}{16}(g_1^6-3g_1^4g_2^2)\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_1]-\frac{3}{16}g_1^4m_1^2(g_1^2-3g_2^2)\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_1]+$$

$$\frac{1}{24}(-6g_1^6-7g_1^4g_2^2+13g_1^2g_2^4)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]+\frac{1}{8}(g_1^6-3g_1^4g_2^2)\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_1]+$$

$$\frac{1}{8}g_1^4m_1^2(g_1^2-3g_2^2)\,\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_1]+\frac{1}{24}(3g_1^6+2g_1^4g_2^2-5g_1^2g_2^4)\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]+$$

$$\frac{1}{24}(9g_1^4g_2^2+2g_1^2g_2^4-11g_2^6)\,\mathrm{LF}_{3,1,-1}[\tilde{\mu},m_2]-\frac{5}{48}(17g_1^2g_2^4-35g_2^6)\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_2]-$$

$$\frac{3}{16}g_2^4m_2^2(g_1^2-3g_2^2)\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_2]-\frac{3}{8}(2g_1^4g_2^2+g_1^2g_2^4-3g_2^6)\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]+$$

$$\frac{1}{8}g_2^4(7g_1^2-17g_2^2)\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_2]+\frac{1}{8}\left(g_2^4m_2^2(g_1^2-3g_2^2)-4g_2^6\tilde{\mu}^2\right)\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_2]+$$

$$\frac{1}{8}(3g_1^4g_2^2+2g_1^2g_2^4-5g_2^6)\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]+(-g_1^4g_2^2+g_1^2g_2^4)\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_1]+$$

$$\frac{3}{4}g_1^2g_2^2\left(m_1m_2(-g_1^2+g_2^2)+4g_2^2\tilde{\mu}^2\right)\mathrm{LF}_{2,2,1,-1}[m_2,\tilde{\mu},m_1]+$$

$$\frac{1}{2}g_1^2g_2^2(g_1^2-g_2^2)\,\mathrm{LF}_{3,2,1,-3}[m_2,\tilde{\mu},m_1]+\frac{1}{2}g_1^2g_2^2\left(m_1m_2(g_1^2-g_2^2)-4g_2^2\tilde{\mu}^2\right)$$

$$\mathrm{LF}_{3,2,1,-2}[m_2,\tilde{\mu},m_1]+\frac{1}{2}g_1^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]-$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}$$

$$\mathrm{LF}_{1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{u}}^{v}]+\frac{1}{2}g_1^2\,\overline{a_d}^{pr}a_d^{sr}\left(\overline{y_d}^{st}y_d^{pt}-\overline{y_u}^{st}y_u^{pt}\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{2}g_1^2y_d^{sr}\,\overline{y_u}^{st}\left(-\overline{a_d}^{pr}a_u^{pt}+\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{1}{2}g_1^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]-\frac{1}{2}g_1^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{2}g_1^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]-\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+\frac{1}{2}g_1^2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]+$$

$$\frac{1}{2}g_1^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]-\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}$$

$$\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]-\frac{1}{2}g_2^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+$$

$$\frac{1}{4}\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$

$$\frac{1}{2}g_2^2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+\frac{1}{4}\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}(-g_1^2+g_2^2)$$

$$\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}]+\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t}]-$$

$$\frac{1}{2}g_2^2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$

$$\frac{1}{4}\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}]+$$

$$\frac{1}{2}\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}(2g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$

$$\frac{3}{4}\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$

$$\frac{1}{2}\,\overline{a_d}^{pr}a_d^{sr}\left(\overline{y_d}^{st}y_d^{pt}(2g_1^2-3g_2^2)-\overline{y_u}^{st}y_u^{pt}(2g_1^2+3g_2^2)\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{4}\,\overline{a_d}^{pr}a_d^{sr}\left(\overline{y_d}^{st}y_d^{pt}(-g_1^2+g_2^2)+\overline{y_u}^{st}y_u^{pt}(g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]-$$

$$\frac{1}{2}y_d^{sr}\,\overline{y_u}^{st}\left(\overline{a_d}^{pr}a_u^{pt}(2g_1^2+3g_2^2)+g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}y_d^{sr}\,\overline{y_u}^{st}\left(\overline{a_d}^{pr}a_u^{pt}(g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}(g_1^2-g_2^2)\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$

$$\frac{1}{2}\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(2g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}a_d^{sr}(g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}]+$$

$$3\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}\,\mathrm{LF}_{1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u}]-$$

$$\frac{1}{4}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}(g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t}]+$$

$$\frac{3}{4}y_d^{sr}\,\overline{y_u}^{st}\left(\overline{a_d}^{pr}a_u^{pt}(g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}(g_1^2-g_2^2)\right)\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t}]+$$

$$\frac{1}{2}\,\overline{a_d}^{pr}a_d^{sr}\left(\overline{y_d}^{st}y_d^{pt}(2g_1^2-3g_2^2)-\overline{y_u}^{st}y_u^{pt}(2g_1^2+3g_2^2)\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{4}\,\overline{a_d}^{pr}a_d^{sr}\left(\overline{y_d}^{st}y_d^{pt}(-g_1^2+g_2^2)+\overline{y_u}^{st}y_u^{pt}(g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]-$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]-$$

$$\frac{1}{4}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}(g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(g_1^2-g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{2}\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(2g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{2}g_2^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{4}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}(7g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{4}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{2}y_d^{sr}\,\overline{y_u}^{st}(2g_1^2-3g_2^2)\left(\overline{a_d}^{pr}a_u^{pt}-\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}\right)\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{2}g_2^2y_d^{sr}\,\overline{y_u}^{st}\left(\overline{a_d}^{pr}a_u^{pt}-\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pt}\right)\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{2}\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-2g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]-$$

$$\frac{3}{2}g_2^2\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{8}g_1^2g_2^2(g_1^2-g_2^2)\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_1,m_2]+\frac{1}{8}(-7g_1^4g_2^2+g_1^2g_2^4)\,\mathrm{LF}_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+$$

$$\frac{3}{8}g_1^2g_2^2\left(-m_1m_2(g_1^2+g_2^2)+8g_2^2\tilde{\mu}^2\right)\mathrm{LF}_{3,1,1,-1}[\tilde{\mu},m_1,m_2]+\frac{1}{4}g_1^4g_2^2\,\mathrm{LF}_{3,2,1,-3}[\tilde{\mu},m_1,m_2]+$$

$$\frac{1}{4}g_1^4g_2^2(m_1^2+2m_1m_2-8\tilde{\mu}^2)\,\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_1,m_2]-2g_1^4g_2^2m_1^2\tilde{\mu}^2\,\mathrm{LF}_{3,2,1,-1}[\tilde{\mu},m_1,m_2]+$$

$$\frac{1}{2}g_1^4g_2^2\,\mathrm{LF}_{4,1,1,-3}[\tilde{\mu},m_1,m_2]+\frac{1}{4}g_1^2g_2^2\left(m_1m_2(g_1^2+g_2^2)-10g_2^2\tilde{\mu}^2\right)\mathrm{LF}_{4,1,1,-2}[\tilde{\mu},m_1,m_2]+$$

$$\frac{1}{4}g_1^2g_2^2(2g_1^2-g_2^2)\,\mathrm{LF}_{3,2,1,-3}[\tilde{\mu},m_2,m_1]+$$

$$\frac{1}{4}g_1^2g_2^2\left(2m_1m_2g_1^2+g_2^2(m_2^2-24\tilde{\mu}^2)\right)\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_2,m_1]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{pt}y_d^{sv}y_d^{ur}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{d}}^{v},m_{\tilde{q}}^{u}]+$$

$$\frac{1}{2}g_1^2\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{a_d}^{st}a_d^{sr}-\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$3\,\overline{y_d}^{st}y_d^{sr}\,\overline{y_u}^{uv}y_u^{pv}\,\overline{a_d}^{pr}a_d^{ut}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u}]+$$

$$3\,\overline{y_d}^{st}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}\left(y_d^{sr}y_d^{ut}a_u^{pv}+y_d^{ur}y_u^{sv}a_d^{pt}\right)\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{u},m_{\tilde{u}}^{v}]+$$

$$3\,\overline{y_d}^{st}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}\left(y_d^{ut}y_u^{pv}a_d^{sr}+y_d^{pt}y_u^{sv}a_d^{ur}\right)\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u}]-$$

$$\frac{1}{2}g_1^2\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+3\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}a_u^{sv}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}\,\overline{y_u}^{uv}y_u^{pv}y_u^{ut}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]-$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$3\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{sv}y_u^{ur}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$3\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{pt}y_u^{ur}a_d^{sv}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,\overline{y_e}^{uv}y_e^{pt}y_e^{sv}y_e^{ur}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{e}}^{v},m_{\tilde{l}}^{u}]+$$

$$\frac{1}{2}g_1^2\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{a_e}^{st}a_e^{sr}-\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]-$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}]-$$

$$\frac{1}{2}g_2^2\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{a_e}^{st}a_e^{sr}+\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+$$

$$\frac{1}{4}\,\overline{a_e}^{pr}a_e^{pt}\left(\overline{a_e}^{st}a_e^{sr}(-g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}(g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}]+$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{sr}\,\overline{a_e}^{pr}a_e^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}]+\frac{1}{2}\,\overline{a_d}^{pr}a_d^{pt}$$

$$\left(\overline{a_d}^{st}a_d^{sr}(2g_1^2-3g_2^2)-\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}(2g_1^2+3g_2^2)\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{4}\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{a_d}^{st}a_d^{sr}(-g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}(g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{2}\left(g_1^2\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}-\overline{a_d}^{pr}a_u^{pt}(2g_1^2+3g_2^2)\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)+\overline{a_d}^{pr}a_u^{pt}(g_1^2+g_2^2)\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{1}{2}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(2g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]-$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\Big(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)+$$

$$\overline{a_d}^{pr}a_d^{sr}\left(\overline{a_u}^{st}a_u^{pt}(g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)\right)\Big)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}]+$$

$$3\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{pv}y_d^{ut}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(-\overline{a_u}^{pr}a_u^{pt}(g_1^2+6g_2^2)+4g_1^2\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(g_2^2\,\overline{a_u}^{pr}a_u^{pt}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(-g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]-$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(\overline{a_u}^{pr}a_u^{pt}(g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(-g_1^2+g_2^2)\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)+\overline{a_d}^{pr}a_u^{pt}(g_1^2+g_2^2)\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(\overline{a_u}^{pr}a_u^{pt}(g_1^2+g_2^2)+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(-g_1^2+g_2^2)\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{sr}\,\overline{a_d}^{pr}a_d^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{2}\left(g_1^2\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}-\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}(2g_1^2+3g_2^2)+g_1^2\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)+\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}(g_1^2+g_2^2)+\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}(g_1^2-g_2^2)\right)\right)$$

$$\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]-\frac{1}{2}g_1^2\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2-g_2^2)\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]-$$

$$\frac{1}{4}g_1^2\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}g_2^2\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(g_1^2-g_2^2)\,\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}(-g_1^2+g_2^2)+\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}(g_1^2+g_2^2)+\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}(g_1^2-g_2^2)\right)\right)$$

$$\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}(-g_1^2+g_2^2)\,\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{2}\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}(-2g_1^2+3g_2^2)\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-$$

$$\frac{3}{2}g_2^2\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]-\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}$$

$$\left(\overline{a_u}^{pr}a_u^{pt}(-4g_1^2+3g_2^2)+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(g_1^2+3g_2^2)\right)\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(g_2^2\,\overline{a_u}^{pr}a_u^{pt}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(-g_1^2+g_2^2)\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\left(2g_2^2\,\overline{a_u}^{pr}a_u^{pt}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{pt}(-g_1^2+g_2^2)\right)\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{2}(2g_1^2-3g_2^2)\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)$$

$$\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+\frac{3}{2}g_2^2$$

$$\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}a_u^{pt}\left(a_d^{sr}\,\overline{a_u}^{st}-\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\right)\right)\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}]+$$

$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}(4g_1^2-3g_2^2)\,\mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$\frac{3}{4}g_2^2\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}\,\mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_d}^{uv}\,\overline{a_d}^{pr}\left(y_d^{sv}y_d^{ur}a_d^{pt}+y_d^{sr}y_d^{ut}a_d^{pv}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u}]+$$

$$3\,\overline{y_u}^{uv}y_u^{sv}\,\overline{a_d}^{pr}\,\overline{a_d}^{st}a_d^{pt}a_d^{ur}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u}]+$$

$$3y_d^{ur}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}a_d^{pt}\left(\overline{a_d}^{st}a_u^{sv}-\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sv}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}\left(y_d^{sr}y_d^{ut}a_u^{pv}+y_d^{ur}y_u^{sv}a_d^{pt}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{pt}y_d^{ur}\,\overline{a_d}^{pr}a_d^{sv}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u}]+$$

$$3\,\overline{y_d}^{uv}\,\overline{a_d}^{pr}\,\overline{a_u}^{st}\left(y_d^{ur}a_d^{sv}a_u^{pt}+y_u^{ut}a_d^{pv}a_d^{sr}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}]+$$

$$3\,\overline{y_d}^{uv}\,\overline{a_d}^{pr}\,\overline{a_u}^{st}\left(y_d^{pv}a_d^{sr}a_u^{ut}+y_d^{sv}a_d^{ur}a_u^{pt}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{pt}y_d^{ur}\,\overline{y_u}^{uv}\,\overline{a_d}^{pr}a_u^{sv}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^2y_d^{sr}\,\overline{y_u}^{st}\,\overline{y_u}^{uv}y_u^{ut}\left(\overline{a_d}^{pr}a_u^{pv}-\tilde{\mu}^2\,\overline{y_d}^{pr}y_u^{pv}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}\,\overline{y_u}^{uv}y_u^{pv}y_u^{ut}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{sv}y_u^{ur}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_u}^{uv}y_u^{sv}y_u^{ur}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}\,\overline{y_d}^{uv}y_d^{pt}y_u^{ur}a_d^{sv}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r}]+$$

$$\tilde{\mu}^2\,\overline{y_e}^{st}\,\overline{y_e}^{uv}\,\overline{a_e}^{pr}\left(y_e^{sv}y_e^{ur}a_e^{pt}+y_e^{sr}y_e^{ut}a_e^{pv}\right)\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{e}}^{v},m_{\tilde{l}}^{p},m_{\tilde{l}}^{u}]+$$

$$\tilde{\mu}^2\,\overline{y_e}^{st}\,\overline{y_e}^{uv}y_e^{pt}y_e^{ur}\,\overline{a_e}^{pr}a_e^{sv}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{v},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{l}}^{u}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{pt}\,\overline{y_u}^{uv}y_u^{ur}\,\overline{a_u}^{pr}a_u^{sv}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^4\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,\overline{y_u}^{uv}y_u^{pt}y_u^{sv}y_u^{ur}\,\mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]+$$

$$3\tilde{\mu}^2\,\overline{y_d}^{uv}y_d^{ur}\,\overline{a_d}^{pr}\,\overline{a_d}^{st}a_d^{pt}a_d^{sv}\,\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u}]+$$

$$3\,\overline{a_d}^{pr}a_d^{pt}\left(\tilde{\mu}^4\,\overline{y_d}^{st}y_d^{ur}\,\overline{y_u}^{uv}y_u^{sv}+\overline{a_d}^{st}a_u^{sv}\left(a_d^{ur}\,\overline{a_u}^{uv}-\tilde{\mu}^2y_d^{ur}\,\overline{y_u}^{uv}\right)\right)$$

$$\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^2\,\overline{y_d}^{uv}y_u^{ut}\,\overline{a_d}^{pr}a_d^{pv}a_d^{sr}\,\overline{a_u}^{st}\,\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{d}}^{v},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t}]+$$

$$3\tilde{\mu}^2\,\overline{y_u}^{uv}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pv}y_u^{ut}+\overline{a_d}^{pr}\left(y_u^{ut}a_d^{sr}\,\overline{a_u}^{st}a_u^{pv}-\tilde{\mu}^2y_d^{ur}\,\overline{y_u}^{st}y_u^{sv}a_u^{pt}\right)\right)$$

$$\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]-$$

$$3\tilde{\mu}^4\,\overline{y_d}^{st}\,\overline{y_u}^{uv}y_u^{sv}y_u^{ur}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r},m_{\tilde{u}}^{v}]+$$

$$\tilde{\mu}^2\,\overline{y_e}^{uv}y_e^{ur}\,\overline{a_e}^{pr}\,\overline{a_e}^{st}a_e^{pt}a_e^{sv}\,\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{e}}^{v},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{l}}^{u}]+$$

$$3\tilde{\mu}^4\,\overline{y_u}^{st}\,\overline{y_u}^{uv}y_u^{sv}y_u^{ur}\,\overline{a_u}^{pr}a_u^{pt}\,\mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{q}}^{u},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{u}}^{v}]\Big)$$